

Assignment 1.3: Event Detection in Missing Label File.

Due date: October 19th, 2026

Part 1:

1. Open excel sheet “Fiber_Photometry_Data_FP_Assign1_3” and inspect your file. Your data is sampled at 0.1-second intervals. Column A represents time into the recording. Column B represents the average DF/F signal for each 0.1-second interval. Column C represents the animal’s sleep-wake state.
2. The DF/F signal has already been preprocessed, so no additional processing is needed.
3. You have data from two mice: Sheet 1 contains data from mouse1, and Sheet 2 contains data from mouse2. Neuronal activity was recorded from the same brain region in both mice.
4. For each mouse, identify peaks within each sleep-wake state (NR, R, and W). Rewrite the code we used in the tutorial 4 as a function called `find_peaks_across_states`. Adjust the threshold so that you can fairly compare peaks across states.

Example Data Frame:

Mouse ID	NR peaks #	R peaks #	W peaks #
Mouse 1	14	56	21
Mouse 2	2	45	44

****Numbers are arbitrary**

Part 2:

What other analyses could you perform on this dataset (for example, transition analysis, as shown in Fig. 1F below or average activity across sleep-wake states in Fig. 1D)? Be creative! You can draw inspiration from outside sources, such as research papers. Choose the analysis you would like to do on your DF/F signal, write the code to carry it out, and interpret your results.

See examples below:

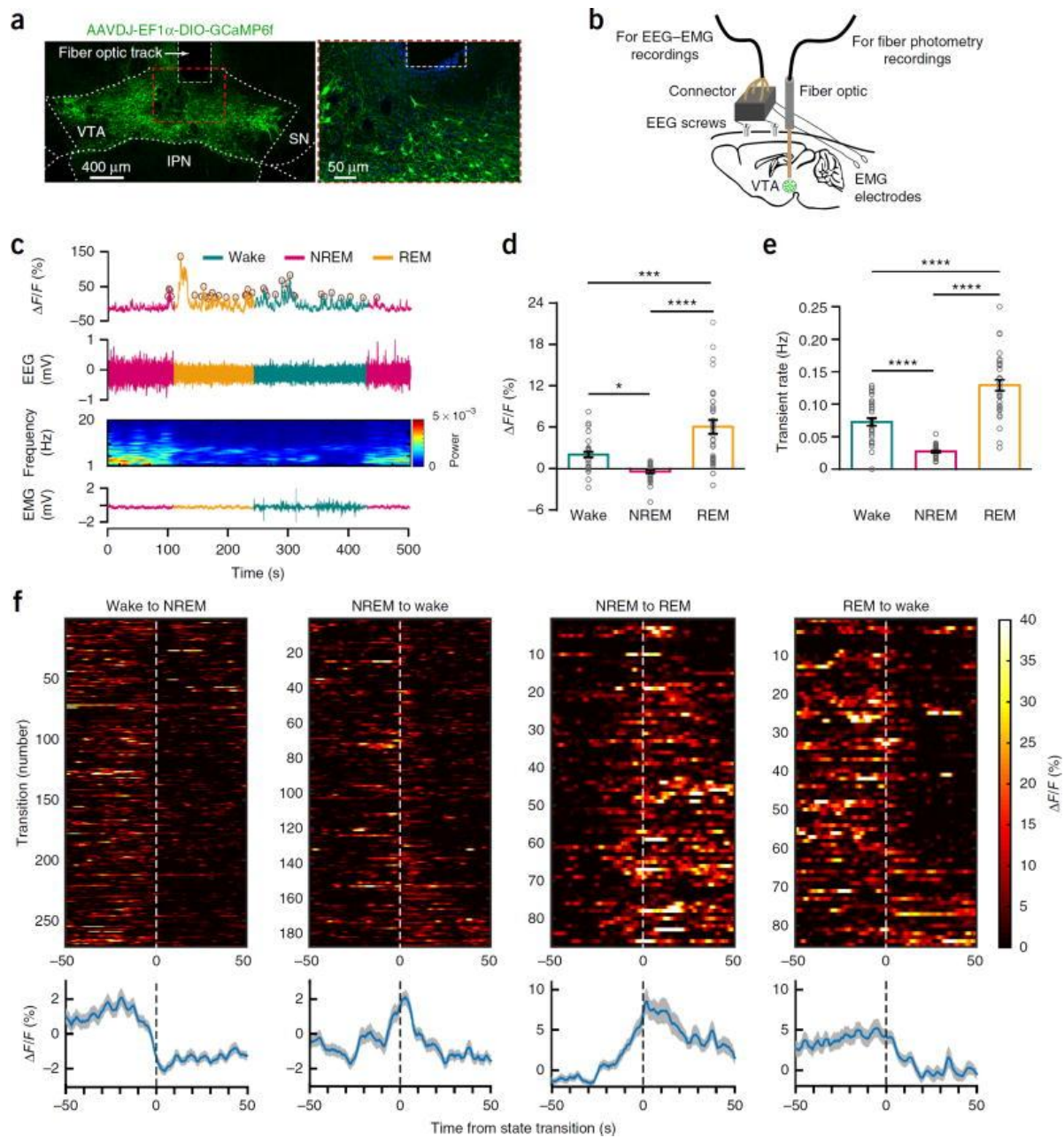


Figure 1. Population activity of Th^+ VTA neurons across sleep–wake states (Eban-Rothschild et al, 2017)

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5519826/>